Real-World Applications & Global Change Projections

Modeling the distribution of biodiversity under global climate change

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FAIR Data and Reproducible Pipelines

Findable, Accessible, Interoperable, and Reusable

Making research more effective: Applying FAIR principles ensures that the data, code, and outputs associated with SDM studies are managed for maximum reuse and impact.

Findable: can others (and machines) find your data and models?

Accessible: once found, how can the data/model be accessed?

Interoperable: can the data be integrated and used with other data or workflows?

Reusable: can the data/model be effectively used in future studies?

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DATA ARTICLE

Bio-ORACLE v3.0. Pushing marine data layers to the CMIP6 Earth System Models of climate change research

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Abstract

Motivation: Impacts of climate change on marine biodiversity are often projected

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journal homepage:

ELSEVIER

Data Article

A global dataset of demosp
records

Data in Brief 48 (2023) 109223

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Data in Brief

journal homepage: www.elsevier.com/locate/dib

ELSEVIER

Data Article

A dataset of cold-water coral distribution

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ABSTRACT

Species distribution data are key for monitoring present and future biodiversity patterns and informing conservation and management strategies. Large biodiversity information facilities often contain spatial and taxonomic errors that reduce the quality of the provided data. Moreover, datasets are frequently shared in varying formats, inhibiting proper integration and interoperability. Here, we provide a quality-controlled dataset of the diversity and distribution of cold-

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SCIENTIFIC DATA

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DATA DESCRIPTOR

A fine-tuned global distribution dataset of marine forests

Jorge Assis[1,4]✉, Eliza Fragkopoulou[1,4], Duarte Frade[1], João Neiva[1], André Oliveira[1], David Abecasis[1], Sylvain Faugeron[2,3] & Ester A. Serrão[1]

Species distribution records are a prerequisite to follow climate-induced range shifts across space and time. However, synthesizing information from various sources such as peer-reviewed literature, herbaria, digital repositories and citizen science initiatives is not only costly and time consuming, but also challenging, as data may contain thematic and taxonomic errors and generally lack standardized

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Data Article

Range map data of marine ecosystem structuring species under global climate change

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ABSTRACT

Data on contemporary and future geographical distributions of marine species are crucial for guiding conservation and

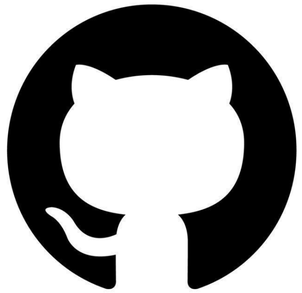


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Open access repositories

Online open access repositories allow researchers to **preserve and share their research outputs, including figures, datasets, images, and videos**.

E.g., Figshare and Zenodo are free to upload content and free to access, in adherence to the principle of open data.



GitHub

Git is an essential tool for **managing complex analytical workflows like SDM**, promoting reproducibility, collaboration, and saving us from losing work or getting lost in different versions of your scripts (strong complement of FAIR Principles).

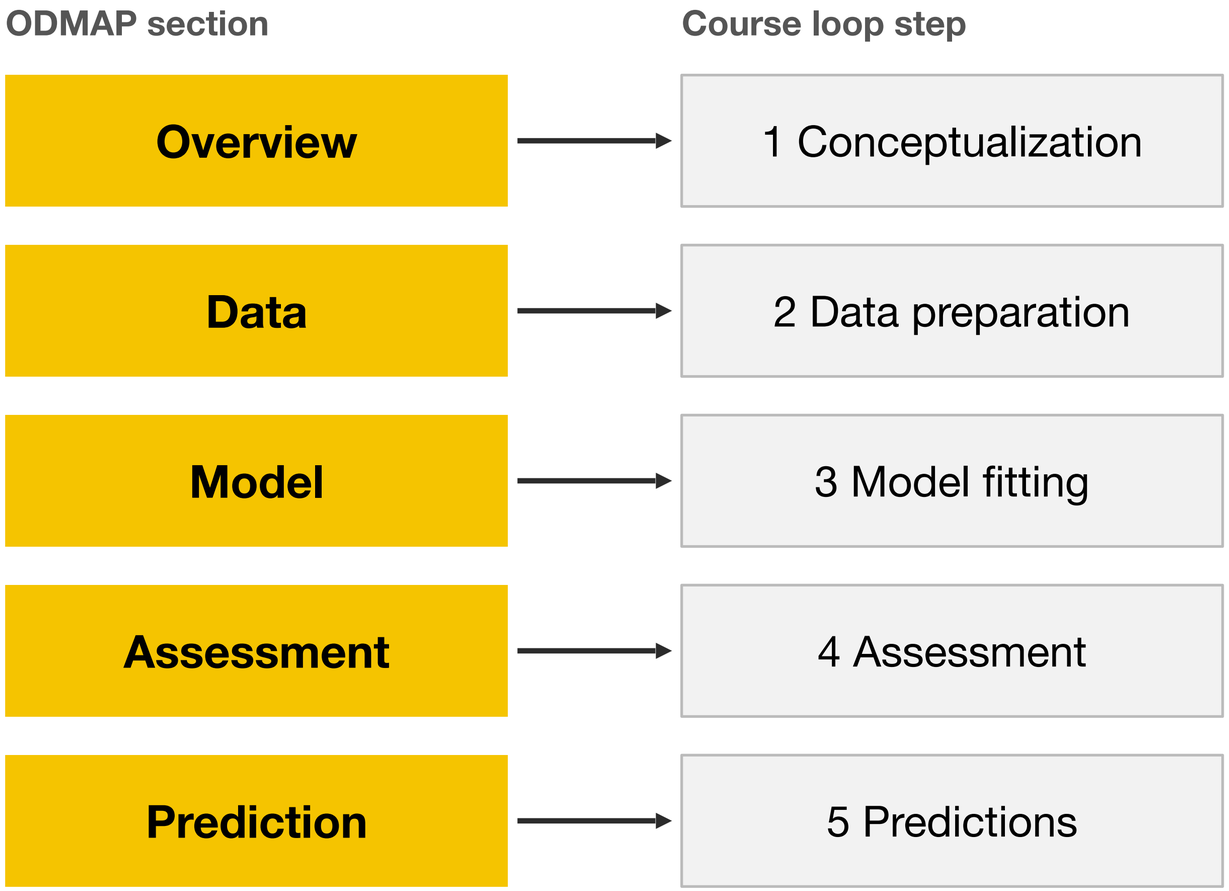
https://github.com/jorgeassis

ODMAP

A standard protocol for reporting SDMs, organised around how modelling decisions serve the objective.

Two uses: a checklist while building the model and a structured format for documentation.

Supports: transparency, reproducibility, peer review and meta-analysis.



Zurell et al., 2020